

Rahul Patwadi

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SKILLS

Languages: Python, TypeScript/JavaScript, C#, SQL

AI/ML: LangGraph, Retrieval-Augmented Generation (RAG), Gemini API, ChromaDB, multi-agent orchestration, prompt engineering

Backend/Frameworks: FastAPI, ASP.NET Web API, REST APIs, Microservices, Pydantic, Full-Stack Development

Frontend: Angular

Data & Messaging: PostgreSQL, Control-M

DevOps: Docker, Kubernetes, Helm, CI/CD, SonarQube, Tenable, Distributed Systems, System Design, APIs, PostgreSQL, Kubernetes, CI/CD

Practices/Tooling: pytest, mypy (strict), ruff, uv, unit/integration testing, ADRs, Agile, Developer Tools, Distributed Systems, System Design

EXPERIENCE

Dell Technologies | *Senior Analyst (Backend / GenAI Engineer)*

Feb 2026 - Aug 2026

- Independently designed, built, and deployed a Python/FastAPI file-delay monitoring service with PostgreSQL and 300-second polling, delivering a production developer tool that has fired verified alerts since February 2026 and cut delay-detection time from hours to near real time.
- Owned CI/CD and deployment across dev, UAT, and production using GitLab, Helm, and Kubernetes, supporting cloud infrastructure for the monitoring service and broader Agentic SRE platform.

Analyst (Backend / GenAI Engineer)

Jun 2025 - Jan 2026

- Collaborated with a small engineering team to build and prompt a 5-agent LLM pipeline for observability, analysis, root-cause identification, solution recommendation, and auto-fix using LangGraph and Llama 3.3, powering automated root-cause analysis for a 37-job production Control-M pipeline polled every 5 minutes.
- Designed a human-in-the-loop approval gate ahead of the auto-fix agent and rebuilt incident-state tracking from a 2-state model to a 3-state model, in progress → pending approval → complete, improving the accuracy of self-healing system workflows and SRE reporting.

Analyst (.NET / Backend Engineer)

Jun 2023 - May 2025

- Contributed to AV Scanner, an existing .NET Web API integrating McAfee antivirus scanning for incoming purchase-order documents; collaborated with teammates to implement synchronous S3 and asynchronous NAS processing flows, migrate the service from a Windows VM to containerized Kubernetes, and raise code quality from 58% to 90%.
- Contributed backend and API work to the Purchase Order Assistant, a FastAPI service that extracted structured data from PDF purchase orders via an ML model, along with its Angular frontend, strengthening full-stack development experience.
- Maintained production microservices by resolving CI/CD, Docker, SonarQube, and Tenable findings and monitoring Kubernetes workloads across the deployment environment.

PROJECTS

NASA Mission Intelligence | github.com/rahul-patwadi/nasa-mission-intelligence

- Built an end-to-end Python and FastAPI Retrieval-Augmented Generation system with an Angular frontend spanning six NASA missions: Apollo, Artemis, ISS, Mars, Hubble, and Voyager; developed an async NTRS document harvester with httpx, pagination, 429 backoff, ITAR/EAR filtering, and deduplication that ingested 539 of 545 records into 86,003 ChromaDB chunks.
- Evaluated retrieval quality with a hand-labeled harness, then truncated embeddings to 768 dimensions, raising mean precision@5 from 0.67 to 0.70 while shrinking the vector store from 1.5GB to 742MB.
- Containerized the full-stack application with Docker, including the Python backend, Angular frontend, and Nginx reverse proxy, while enforcing engineering best practices with uv, ruff, mypy --strict, and a pytest suite.

EDUCATION

San José State University

M.S., Engineering (Specialization: AI Systems Engineering)

Aug 2026 - May 2028

San Jose, CA

JSS Academy of Technical Education (VTU)

B.E., Electronics and Communication Engineering

2018 - 2022

Bangalore, India